PRODUCT AND QUOTIENT RULES

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Name: Solutions

In each of the problems, use the derivative rules to differentiate the given function. Use proper notation and simplify your final answers. In some cases it might be advantageous to simplify/rewrite first. Do not use rules found in later sections.

1. $f(x) = \left(1 + \sqrt{x}\right) x^3$

$= x^3 + x^{3+\frac{1}{2}}$

$= x^3 + x^{7/2}$

$\boxed{f'(x) = 3x^2 + 7/2\, x^{5/2}}$

2. $g(t) = \left(\frac{2}{t} + t^5\right)(t^3 + 1)$

$= 2t^2 + \frac{2}{t} + t^8 + t^5$

$= 2t^2 + 2t^{-1} + t^8 + t^5$

$g'(t) = 4t - 2t^{-2} + 8t^7 + 5t^4$

$= \boxed{8t^7 + 5t^4 + 4t - 2/t^2}$

3. $h(y) = \frac{1}{y^3 + 2y + 1}$

$h'(y) = \frac{\frac{d}{dy}(1)(y^3+2y+1) - (1)\frac{d}{dy}(y^3+2y+1)}{(y^3+2y+1)^2}$

$= \frac{0 - (3y^2+2y)}{(y^3+2y+1)^2}$

$= \boxed{\frac{-3y^2-2y}{(y^3+2y+1)^2}}$

4. $y = \frac{1}{x + \sqrt{x}}$

$f(x) = 1, \quad f'(x) = 0, \quad g(x) = x + \sqrt{x} = x + x^{1/2}, \quad g'(x) = 1 + \frac{1}{2}x^{-1/2} = 1 + \frac{1}{2\sqrt{x}}$

$\frac{dy}{dx} = \frac{d}{dx}\frac{f(x)}{g(x)} = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2} = \frac{0(x+\sqrt{x}) - (1)(1+\frac{1}{2\sqrt{x}})}{(x+\sqrt{x})^2}\frac{2\sqrt{x}}{2\sqrt{x}}$

$= \boxed{\frac{-2\sqrt{x} - 1}{2\sqrt{x}(x+\sqrt{x})^2}}$

5. $g(z) = \frac{z^2 + 1}{z^3 - 5}$

$g'(z) = \frac{\frac{d}{dz}(z^2+1)(z^3-5) - (z^2+1)\frac{d}{dz}(z^3-5)}{(z^3-5)^2}$

$= \frac{(2z+0)(z^3-5) - (z^2+1)(3z^2-0)}{(z^3-5)^2}$

$= \frac{2z^4 - 10z - 3z^4 - 3z^2}{(z^3-5)^2} = \boxed{\frac{-z^4 - 3z^2 - 10z}{(z^3-5)^2}}$

6. $y = \frac{\sqrt{x}}{x^3 + 1}$

$\frac{dy}{dx} = \frac{\frac{d}{dx}(x^{1/2})(x^3+1) - \sqrt{x}\,\frac{d}{dx}(x^3+1)}{(x^3+1)^2}$

$= \frac{(\frac{1}{2}x^{-1/2})(x^3+1) - \sqrt{x}(3x^2)}{(x^3+1)^2}$

$= \frac{\frac{1}{2}x^{5/2} + \frac{1}{2}x^{-1/2} - 3x^{5/2}}{(x^3+1)^2} = \frac{-5/2\, x^{5/2} + \frac{1}{2}x^{-1/2}}{(x^3+1)^2}\left(\frac{2\sqrt{x}}{2\sqrt{x}}\right)$

$= \boxed{\frac{-5x^3 + 1}{2\sqrt{x}(x^3+1)^2}}$

7. $z = \dfrac{t^2}{(t-4)(2-t^3)}$

$$\frac{dz}{dt} = \frac{2t(t-4)(2-t^3) - t^2(2-t^3 + (t-4)(-3t^2))}{(t-4)^2(2-t^3)^2}$$

$$= \frac{4t^2 - 2t^5 - 16t + 8t^4 - 2t^2 + t^5 + 3t^5 - 12t^4}{(t-4)^2(2-t^3)^2}$$

$$= \frac{(-2+1+3)t^5 + (8-12)t^4 + (4-2)t^2 - 16t}{(t-4)^2(2-t^3)^2} = \boxed{\frac{2t^5 - 4t^4 + 2t^2 - 16t}{(t-4)^2(2-t^3)^2}}$$

8. $h(x) = \dfrac{(x^3+1)\sqrt{x}}{x^2}$

$$h(x) = \frac{(x^3+1)x^{1/2}}{x^2} = \frac{x^{7/2} + x^{1/2}}{x^{4/2}} = \frac{x^{7/2}}{x^{4/2}} + \frac{x^{1/2}}{x^{4/2}}$$

$$= x^{7/2 - 4/2} + x^{1/2 - 4/2} = x^{3/2} + x^{-3/2}$$

$$h'(x) = 3/2\, x^{3/2 - 2/2} + (-3/2)x^{-3/2 - 2/2}$$

$$= \boxed{3/2\, x^{1/2} - 3/2\, x^{-5/2}}$$

9. Let $f(x) = g(x)h(x)$, $g(10) = -4$, $h(10) = 560$, $g'(10) = 0$, $h'(10) = 4$. Find $f'(10)$.

$$f'(10) = g'(10)h(10) + g(10)h'(10)$$

$$= 0(560) + (-4)(4) = \boxed{-16.}$$

10. Let $z(-3) = 6$, $z'(-3) = 15$, and

$$y(x) = \frac{z(x)}{1+x^2}.$$

Find $y'(-3)$.

$$y'(x) = \frac{z'(x)(1+x^2) - z(x)(2x)}{(1+x^2)^2}$$

$$y'(-3) = \frac{z'(3)(1+9) - z(3)(-6)}{(1+9)^2} = \frac{15(10) + 6(6)}{10^2} = \frac{150+36}{100} = \boxed{\frac{186}{100} = 1.86.}$$

Use the derivatives

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin(x) = \cos(x) \text{ and } \frac{\mathrm{d}}{\mathrm{d}x}\cos(x) = -\sin(x)$$

and the trigonometric identities

- $\tan(x) = \dfrac{\sin(x)}{\cos(x)}$
- $\cot(x) = \dfrac{1}{\tan(x)} = \dfrac{\cos(x)}{\sin(x)}$
- $\sec(x) = \dfrac{1}{\cos(x)}$
- $\csc(x) = \dfrac{1}{\sin(x)}$

to compute the given derivatives.

11. $\dfrac{\mathrm{d}}{\mathrm{d}x}\tan(x)$

$$\frac{d}{dx}\frac{\sin(x)}{\cos(x)} = \frac{\cos(x)\cos(x) - \sin(x)(-\sin(x))}{\cos^2(x)}$$

$$= \frac{\cos^2(x)+\sin^2(x)}{\cos^2(x)}$$

$$= \frac{1}{\cos^2(x)}$$

$$= \boxed{\sec^2(x).}$$

12. $\dfrac{\mathrm{d}}{\mathrm{d}x}\cot(x)$

$$\frac{d}{dx}\cot(x) = \frac{d}{dx}\left(\frac{1}{\tan(x)}\right) = \frac{\frac{d}{dx}(1)\tan(x) - 1\frac{d}{dx}\tan(x)}{\tan^2(x)} = -\frac{\sec^2(x)}{\tan^2(x)}$$

$$= -\sec^2(x)\cot^2(x) = \frac{-1}{\cos^2(x)}\frac{\cos^2(x)}{\sin^2(x)} = \frac{-1}{\sin^2(x)}$$

$$= \boxed{-\csc^2(x)}$$

13. $\frac{\mathrm{d}}{\mathrm{d}x}\sec(x)$

$$\frac{d}{dx}\sec(x) = \frac{d}{dx}\frac{1}{\cos(x)} = \frac{0-(-\sin(x))}{\cos^2(x)} = \frac{1}{\cos(x)}\frac{\sin(x)}{\cos(x)} = \boxed{\sec(x)\tan(x)}$$

14. $\frac{\mathrm{d}}{\mathrm{d}x}\csc(x)$

$$\frac{d}{dx}\csc(x) = \frac{d}{dx}\frac{1}{\sin(x)} = \frac{0-\cos(x)}{\sin^2(x)} = \frac{-1}{\sin(x)}\frac{\cos(x)}{\sin(x)} = \boxed{-\csc(x)\cot(x)}$$